

From modelling electricity markets to system planning

So far, we only looked at the **short-term perspective**, i.e. how to make the best use of the existing electricity system infrastructure.

Degrees of freedom / decision variables:

- | | |
|------------------------------|-----------------------------------|
| 1 Dispatch of power plants | 4 Investment in new generation |
| 2 Operation of storage units | 5 Investment in new storage units |
| 3 Curtailment of renewables | 6 Transmission reinforcement |

Now, we will also look at the **long-term perspective**. We will optimise investment in the capacities of generators, storage and transmission by promoting the capacities, like G and F , to decision variables in the optimisation problem.

Electricity system **operation** as optimisation problem

Decision variables (all operational) are shown in **red**. **Capacity expansion** is an extension!

$$\min_{g,e,f} \sum_{i,s,t} o_{i,s} \mathbf{g}_{i,s,t} \quad \text{s.t.}$$

$$d_{i,t} = \sum_s \mathbf{g}_{i,s,t} - \sum_{\ell} K_{i\ell} \mathbf{f}_{\ell,t} \quad \text{energy balance}$$

$$0 \leq \mathbf{g}_{i,s,t} \leq \hat{g}_{i,s,t} G_{i,s} \quad \text{generator limits}$$

$$0 \leq \mathbf{g}_{i,s,t,\text{discharge}} \leq G_{i,s,\text{discharge}} \quad \text{storage discharge limits}$$

$$0 \leq \mathbf{g}_{i,s,t,\text{charge}} \leq G_{i,s,\text{charge}} \quad \text{storage charge limits}$$

$$0 \leq \mathbf{e}_{i,s,t} \leq E_{i,rs} \quad \text{storage energy limits}$$

$$\mathbf{e}_{i,r,t} = \eta_{i,s,t}^0 \mathbf{e}_{i,s,t-1} + \eta_s^1 \mathbf{g}_{i,s,t,\text{charge}} - \frac{1}{\eta_s^2} \mathbf{g}_{i,s,t,\text{discharge}} \quad \text{storage consistency}$$

$$-F_{\ell} \leq \mathbf{f}_{\ell,t} \leq F_{\ell} \quad \text{line limits}$$

$$0 = \sum_{\ell} C_{\ell c} x_{\ell} \mathbf{f}_{\ell,t} \quad \text{KVL}$$

Electricity system **planning** as optimisation problem

In addition to previous **operational decision variables**, we convert the parameters for capacities into **investment decision variables** with **new associated parameters**.

$$\min_{g,e,f,G,E,F} \sum_{i,s,t} w_t o_s g_{i,s,t} + \sum_{i,s} c_s G_{i,s} + c_{r,\text{dis/charge}} G_{i,r,\text{dis/charge}} + c_r E_{i,r} + c_\ell F_\ell \quad \text{s.t.}$$

$$d_{i,t} = \sum_s g_{i,s,t} - \sum_\ell K_{i\ell} f_{\ell,t} \quad \text{energy balance}$$

$$0 \leq g_{i,s,t} \leq \hat{g}_{i,s,t} G_{i,s} \quad \text{generator limits}$$

$$0 \leq g_{i,r,t,\text{discharge}} \leq G_{i,r,\text{discharge}} \quad \text{storage discharge limits}$$

$$0 \leq g_{i,r,t,\text{charge}} \leq G_{i,r,\text{charge}} \quad \text{storage charge limits}$$

$$0 \leq e_{i,r,t} \leq E_r \quad \text{storage energy limits}$$

$$e_{i,r,t} = \eta_{i,r,t}^0 e_{i,r,t-1} + \eta_r^1 g_{i,r,t,\text{charge}} - \frac{1}{\eta_r^2} g_{i,r,t,\text{discharge}} \quad \text{storage consistency}$$

$$-F_\ell \leq f_{\ell,t} \leq F_\ell \quad \text{line limits}$$

$$0 = \sum_\ell C_{\ell c} x_\ell f_{\ell,t} \quad \text{KVL}$$

Changes in objective function with capacity expansion.

$$\min_{g,e,f,G,E,F} \sum_{i,s,t} w_t o_s g_{i,s,t} + \sum_{i,s} c_s G_{i,s} + c_{r,\text{dis/charge}} G_{i,r,\text{dis/charge}} + c_r E_{i,r} + c_\ell F_\ell$$

- We have **marginal cost** o_s for each generator (STMGC, €/MWh),
- we now also have **specific capital costs** $c_\star$ (e.g. building new power plant, €/MW),
- and a weighting w_t denoting how much time a time step t represents.
- Altogether, this objective function minimises **total system costs**.

Units of objective function with capacity expansion

$$\min \quad \overbrace{\sum_{i,s,t} h \cdot \frac{\text{€}}{\text{MWh}} \cdot \text{MW}}^{\text{operational costs}} + \overbrace{\sum_{i,s} \frac{\text{€}}{\text{MW}} \cdot \text{MW} + \frac{\text{€}}{\text{MWh}} \cdot \text{MWh}}^{\text{investment costs}}$$

- For a fair comparison between operational costs and overnight investment costs, we need to align the terms such that they **represent the same time frame**.
- **One common option: yearly system costs (€/a)**
 - **operational**: one year of representative operational conditions ($\sum_t w_t = 8760\text{h}$)
 - **investment**: annualised investment costs ($c_\star$ in $\text{€/MW/a} = \text{€/MW}/8760\text{h}$)

The value of money depends on time

Question: What would you prefer: 1000 € today, or 1000 € in 3 years?

Suppose 1000 € today can be invested in the bank with an interest rate of 5%.

After 3 years you would have:

$$1000 \cdot (1 + 0.05)^3 = 1157.63\text{€}$$

Answer: Best to take the money today and use the opportunity to invest!

Present Value

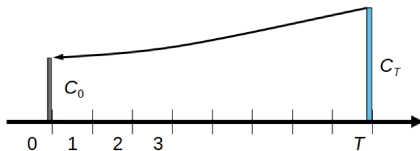
To allow comparison between income and costs in different years, we need to agree on a particular point in time to evaluate the cash flows; e.g. today's value or **present value**.

For an **interest rate** r , we multiply the cash flow in t years by the discount factor

$$\frac{1}{(1+r)^t}$$

to calculate the present value. We have **discounted** the future cash flow.

Future income or costs are **worth less** from today's point of view (if $r > 0$).

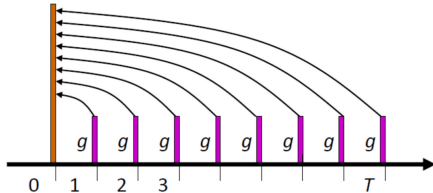


The present value factor for recurring income or costs

For a series of cash flows that are the same every year, we can derive a formula to calculate the sum of discounted cash flows, the **present value factor**:

$$PVF(r, T) = \sum_{t=1}^T \frac{1}{(1+r)^t} = \frac{1}{r} \left[1 - \frac{1}{(1+r)^T} \right]$$

This factor brings a series of **equal annual cash flows** to the **present value**.



Example: $PVF(5\%, 20 \text{ years}) = 12.46$

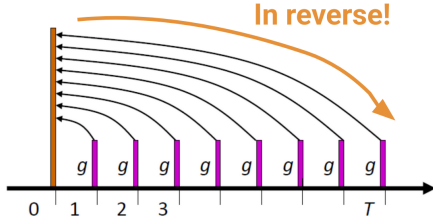
A wind turbine that returns a cash flow of 100,000 €/year for 20 years, may not cost more than 1.246 M€ today, even though the total cash flow is 2 M€.

The annuity factor – opposite of the present value factor

Vice versa, we can use the present value factor to convert an **initial investment** to **annual payments** to compare them with the cash flow in a year:

$$a(r, T) = \frac{1}{PVF(r, T)} = \frac{r}{1 - (1 + r)^{-T}}$$

It works similar to a mortgage for a house.



Example: $PVF(5\%, 20 \text{ years}) = 12.46$

The upfront investment of a wind turbine is 1 M€. Distributed over its 20 year lifetime, this corresponds to an annuity of 80,257 €/a (~ annual cost). This is the required revenue needed for a profitable investment.